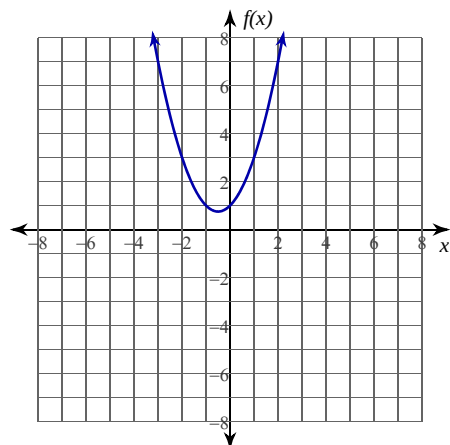


Instantaneous Rates of Change

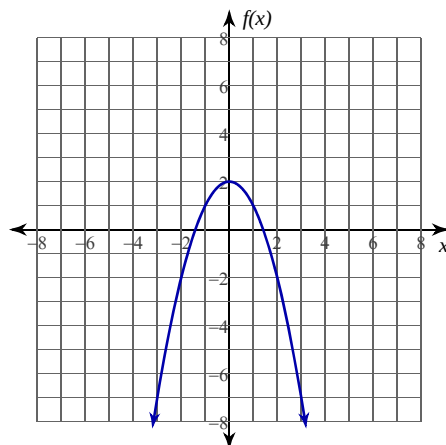
Date _____ Period _____

For each problem, find the instantaneous rate of change of the function at the given value.

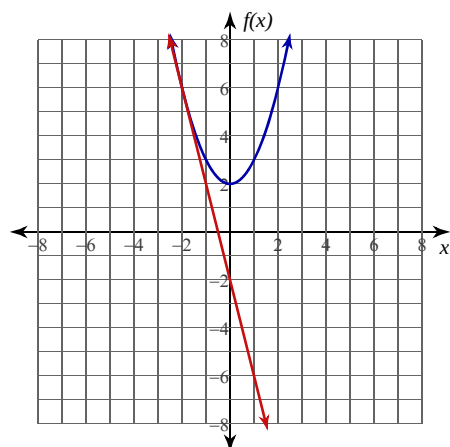
1) $f(x) = x^2 + x + 1$; -2

 -3

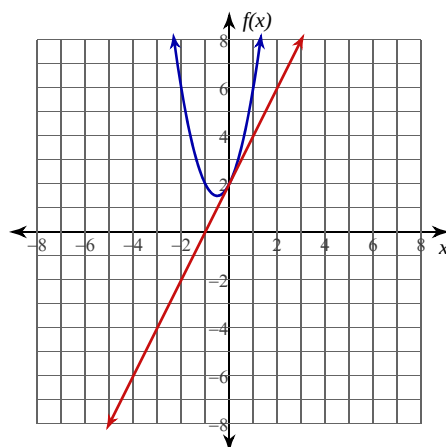
2) $f(x) = -x^2 + 2$; -2

 4 **For each problem, find the equation of the tangent line to the function at the given point.**

3) $f(x) = x^2 + 2$; $(-2, 6)$

 $y = -4x - 2$

4) $f(x) = 2x^2 + 2x + 2$; $(0, 2)$

 $y = 2x + 2$ **Critical thinking question:**5) Look back to problem 1. At what value of x is the derivative 0?The derivative is 0 at $x = -\frac{1}{2}$.